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NUMERICAL SOLUTION OF THE STATIONARY NAVIER–STOKES EQUATIONS USING A MULTILEVEL FINITE ELEMENT METHOD*

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Abstract. The numerical implementation of a multilevel finite element method for the steady-state Navier–Stokes equations is considered. The multilevel method proposed here for the Navier–Stokes equations is a multiscale method in which the full nonlinear Navier–Stokes equations are only solved on a single coarse grid; subsequent approximations are generated on a succession of refined grids by solving a linearized Navier–Stokes problem. Two numerical examples are considered: the first is an example for which an exact solution is known and the second is the driven cavity problem. We demonstrate numerically that for an appropriate choice of grids, a two- or three-level finite element method is significantly more efficient than the standard one-level finite element method.

Key words. multilevel, Navier–Stokes equation, finite elements

AMS subject classifications. 65N30, 35J60

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1. Introduction.

In this paper we consider the numerical implementation of a multilevel finite element method for the steady-state Navier–Stokes equations which was proposed by Layton [6] and Layton and Leferink [7]. The discretization of the Navier–Stokes equations using standard finite element methods has been considered by several authors; for example, see [3], [4], [8], and [9]. In the multilevel finite element method described here we use a sequence of grids and solve the full nonlinear equations only on the coarsest mesh. Subsequent iterations are formed by solving the Navier–Stokes equations linearized about the solution on the previous mesh. In [6] an error estimate is provided which gives guidelines for choosing the grid spacings. These error estimates are asymptotic in nature. Thus it is important to test the algorithms to see if the asymptotic convergence rates are obtained for computational feasible meshes and to check that the theoretical advantages of the algorithms are seen in practical tests. We present two tests: the first is a nonphysical one, but one for which an exact solution is known, while the second is the driven cavity problem. In our tests we consistently observe that when we compare the work done in the solution of the linear and nonlinear systems a two-level finite element method is three to four times more efficient than the standard one-level finite element method.

This paper is organized into three sections. In the remainder of this section we establish the notation that will be used throughout the work and present the standard weak formulation of the steady-state Navier–Stokes equations which we will use. In the second section the multilevel finite element method is described and the error estimate established in [6] is presented. The last section describes the implementation of the algorithm and the computational results of two experiments.

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1.1. Notation. Let Ω be an open, bounded region in R^2 or R^3 with a Lipschitz continuous boundary Γ. Throughout this paper we will use the standard notation for Sobolev spaces (see [1]). Specifically $H^r(\Omega)$, where r is an integer greater than zero, will denote the Sobolev space of real-valued functions with square integrable derivatives of order up to r equipped with the usual norm which we denote by $\|\cdot\|_r$. We will denote $H^0(\Omega)$ by $L^2(\Omega)$ and the standard L^2 inner product by $(\cdot,\cdot)$. Also $\mathbf{H}^r(\Omega)$ will denote the space of vector-valued functions each of whose n components belong to $H^r(\Omega)$, and the dual space of $\mathbf{H}^r(\Omega)$ will be denoted by $\mathbf{H}^{-r}(\Omega)$. The trace space of functions defined on the boundary of Ω will be denoted by $H^s(\Gamma)$. Of particular interest to us will be the constrained spaces

$$L_0^2(\Omega) = \left\{ \psi \in L^2(\Omega) \,:\, \int_\Omega \psi \, d\Omega = 0 \right\}$$

and

$$\mathbf{H}_0^1(\Omega) = \{ \mathbf{v} \in \mathbf{H}^1(\Omega) \,:\, \mathbf{v} = 0 \text{ on } \Gamma \} .$$

We will also make use of the seminorm $|\mathbf{u}|_1 = \|\mathrm{grad}\,\mathbf{u}\|_0$ for $\mathbf{u} \in \mathbf{H}^1(\Omega)$.

1.2. The stationary Navier–Stokes equations. We are interested in approximating the solution of the steady-state Navier–Stokes equations written in primitive variable formulation in terms of the velocity $\mathbf{u}$ and the pressure p. In particular, we consider

$$\begin{aligned} -\frac{1}{\mathcal{R}e}\Delta\mathbf{u} + \mathbf{u}\cdot\mathrm{grad}\,\mathbf{u} + \mathrm{grad}\,p &= \mathbf{f} \quad \text{in } \Omega, \\ \mathrm{div}\,\mathbf{u} &= g \quad \text{in } \Omega, \\ \mathbf{u} &= \mathbf{q} \quad \text{on } \Gamma\,, \end{aligned} \tag{1}$$

where $\mathcal{R}e$ is the Reynolds number and $\mathbf{f} \in \mathbf{H}^{-1}(\Omega)$ is the given body force per unit mass; also $g \in L_0^2(\Omega)$ and $\mathbf{q} \in \mathbf{H}^{1/2}(\Gamma)$ are given functions such that

$$\int_\Omega g \, d\Omega = \int_\Gamma \mathbf{q}\cdot\mathbf{n}\, d\Gamma = 0\,.$$

In the following exposition we will represent the inverse Reynolds number by ν. Note that we have written the system in the general form where both the partial differential equations and the boundary conditions are inhomogeneous. If $g = 0$ then we have the incompressible Navier–Stokes equations and if $\mathbf{q} = \mathbf{0}$ then we are imposing a standard no-slip boundary condition.

We consider the following standard weak formulation of (1); see [3], [4]. Seek $(\mathbf{u}, p) \in \mathbf{H}^1(\Omega) \times L_0^2(\Omega)$ such that

$$\begin{aligned} \nu\, a_0(\mathbf{u},\mathbf{v}) + a_1(\mathbf{u};\mathbf{u},\mathbf{v}) + b(\mathbf{v},p) &= (\mathbf{f},\mathbf{v}) \quad \forall\, \mathbf{v} \in \mathbf{H}_0^1(\Omega)\,, \\ b(\mathbf{u},\psi) &= -(g,\psi) \quad \forall\, \psi \in L_0^2(\Omega)\,, \end{aligned} \tag{2}$$

where

$$a_0(\mathbf{u},\mathbf{v}) = \int_\Omega \mathrm{grad}\,\mathbf{u}\cdot\mathrm{grad}\,\mathbf{v}\, d\Omega - \frac{1}{2}\int_\Omega g\mathbf{u}\cdot\mathbf{v}\, d\Omega \quad \forall\, \mathbf{u},\mathbf{v} \in \mathbf{H}^1(\Omega)\,, \tag{3}$$

$$\begin{aligned} a_1(\mathbf{u};\mathbf{w},\mathbf{v}) &= \frac{1}{2}\left(\int_\Omega \mathbf{u}\cdot\mathrm{grad}\,\mathbf{w}\cdot\mathbf{v}\, d\Omega - \int_\Omega \mathbf{u}\cdot\mathrm{grad}\,\mathbf{v}\cdot\mathbf{w}\, d\Omega \right) \\ &\quad \forall\, \mathbf{u},\mathbf{v},\mathbf{w} \in \mathbf{H}^1(\Omega)\,, \end{aligned} \tag{4}$$

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and

$$b(\mathbf{u},\psi) = -\int_\Omega \psi \operatorname{div} \mathbf{u}\, d\Omega \quad \forall\, \mathbf{u} \in \mathbf{H}^1(\Omega), \quad \psi \in L^2(\Omega)\,. \tag{5}$$

This weak formulation reduces to the standard one used in the case of homogeneous boundary conditions and $g = 0$; see [3].

Continuity of the forms $a_0(\cdot,\cdot)$, $a_1(\cdot,\cdot,\cdot)$, and $b(\cdot,\cdot)$, as well as the standard stability conditions, can be demonstrated. These conditions, along with an assumption on the data guarantee existence and uniqueness of a solution $(\mathbf{u}, p)$ of (2). For example, if $g = 0$ and $\mathbf{q} = 0$ then the necessary assumption on the data is $\nu^2 - N\|\mathbf{f}\|_{-1} > 0$, where N is the continuity constant for the trilinear form $a_1(\cdot,\cdot,\cdot)$. See [3], [4] for details.

1.3. Finite element discretization. For the standard finite element discretization of (2) we choose finite element subspaces $\mathbf{V}^h \subset \mathbf{H}^1(\Omega)$ and $S^h \subset L^2_0(\Omega)$. We then calculate $\mathbf{u}^h \in \mathbf{V}^h$ and $p^h \in S^h$ satisfying

$$\begin{aligned} a_0(\mathbf{u}^h,\mathbf{v}^h) + a_1(\mathbf{u}^h,\mathbf{u}^h,\mathbf{v}^h) + b(p^h,\mathbf{v}^h) &= (\mathbf{f},\mathbf{v}^h) \quad \forall \mathbf{v}^h \in \mathbf{V}^h_0, \\ b(\mathbf{u}^h,\psi^h) &= -(g,\psi^h) \quad \forall \psi^h \in S^h, \\ \mathbf{u}^h &= \mathbf{q}^h \quad \text{ on } \Gamma\,, \end{aligned} \tag{6}$$

where $\mathbf{q}^h$ is an approximation to $\mathbf{q}$ on Γ and $\mathbf{V}^h_0 = \mathbf{V}^h \cap \mathbf{H}^1_0(\Omega)$. Analogous to the continuous problem (2), one can prove existence and uniqueness for the solution of the discrete problem (6). Of course, the stability conditions must hold over the finite element spaces; again see [4] for details. In addition, optimal error estimates can be derived; for the homogeneous problem see [3], [8] and for the inhomogeneous problem see [4].

Once the finite element spaces are prescribed, the discrete problem (6) reduces to solving a system of nonlinear algebraic equations which has a Jacobian that is large, sparse, and banded. Various iterative methods to solve the nonlinear problem (6) are analyzed in [3] for homogeneous boundary conditions and [4] for the inhomogeneous case. For example, a standard approach is to use Newton's method to linearize (6) for a fixed mesh size h. In this case we choose a $\mathbf{u}^h_0$ such that $\mathbf{u}^h_0|_\Gamma = \mathbf{q}^h$ and generate a sequence $\{\mathbf{u}^h_\ell, p^h_\ell\}$, $\ell \geq 1$, by solving the linear problem

$$\begin{aligned} a_0(\mathbf{u}^h_\ell,\mathbf{v}^h) + a_1(\mathbf{u}^h_{\ell-1},\mathbf{u}^h_\ell,\mathbf{v}^h) + a_1(\mathbf{u}^h_\ell,\mathbf{u}^h_{\ell-1},\mathbf{v}^h) + b(p^h_\ell,\mathbf{v}^h) \\ = (\mathbf{f},\mathbf{v}^h) + a_1(\mathbf{u}^h_{\ell-1}\mathbf{u}^h_{\ell-1},\mathbf{v}^h) \quad \forall \mathbf{v}^h \in \mathbf{V}^h_0, \end{aligned} \tag{7}$$

$$b(\mathbf{u}^h_\ell,\psi^h) = -(g,\psi^h) \quad \forall \psi^h \in S^h\,.$$

Since only one fixed mesh spacing is used, we will refer to this approach as a one-level, Newton finite element method or simply a *one-level method*. In the next section we will describe a multilevel, Newton finite element discretization or simply a *multilevel discretization* for solving the discrete problem (6) which will use a Newton linearization.

2. A multilevel method for the Navier–Stokes equations. In this section we describe a multilevel discretization for approximating the solution of the steady-state Navier–Stokes equations. In this method we solve the nonlinear system (6) only on one coarse grid and then solve linearized problems such as (7) on a set of finer grids. We state an error estimate proved in [6] which suggests a choice of grids which will give optimal convergence.

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2.1. The algorithm. We consider a sequence of mesh spacings, h_j, $j = 0, 1, \ldots, J$, such that $h_0 > h_1 > \cdots > h_J$ and let $\mathbf{V}^{h_j}(\Omega)$, $S^{h_j}(\Omega)$ be the finite element spaces corresponding to each mesh size h_j where

$$\mathbf{V}^{h_0} \subset \mathbf{V}^{h_1} \subset \cdots \subset \mathbf{V}^{h_J} \subset \mathbf{H}^1(\Omega)$$

and

$$S^{h_0} \subset S^{h_1} \subset \cdots \subset S^{h_J} \subset L_0^2(\Omega).$$

It is not necessary that the spaces $\mathbf{V}^{h_j}$ and S^{h_j} be nested; however, in our tests we will choose them in this manner. We will also need the spaces $\mathbf{V}_0^{h_j} = \mathbf{V}^{h_j} \cap \mathbf{H}_0^1(\Omega)$ for $j = 1, 2, \ldots, J$. As before, we choose the finite element spaces so that $(\mathbf{V}^{h_j}, S^{h_j})$ will satisfy the inf-sup condition; see [5].

Suppose we wish to approximate a solution of the Navier–Stokes equations in $(\mathbf{V}^{h_j}, S^{h_j})$. An approximate solution can be calculated using (7), i.e., generating a sufficiently good initial guess followed by iteratively applying Newton's method until convergence is attained. At each Newton step this procedure involves reassembly of the linearized system and residual and the solution of the resulting linear system. The idea of the multilevel algorithm is to generate a sequence of iterates $\{\mathbf{u}^{h_j}, p^{h_j}\}$, $j = 0, 1, 2, \ldots, J$, which approximate the solution to the nonlinear problem (2) in $(\mathbf{V}^{h_j}, S^{h_j})$. To obtain $(\mathbf{u}^{h_0}, p^{h_0})$ we solve the nonlinear problem (6) on the coarsest mesh h_0. To obtain $(\mathbf{u}^{h_1}, p^{h_1})$ we solve a discrete linearized Navier–Stokes problem which is linearized about $(\mathbf{u}^{h_0}, p^{h_0})$; to obtain the solution at the next mesh level h_2 we solve (6) linearized about $(\mathbf{u}^{h_1}, p^{h_1})$. In this manner the full nonlinear problem is only solved on the coarsest mesh and on each refined mesh one larger, but the linear system is solved.

A more general algorithm would be to solve S linear problems on each mesh level. For example, to obtain $(\mathbf{u}^{h_1}, p^{h_1})$ we generate a sequence $\{(\mathbf{u}_s^{h_1}, p_s^{h_1})\}$, $s = 0, 1, \ldots, \mathcal{S}$, such that $(\mathbf{u}_0^{h_1}, p_0^{h_1}) = (\mathbf{u}^{h_0}, p^{h_0})$ and $(\mathbf{u}_s^{h_1}, p_s^{h_1})$ is generated by solving (6) linearized about $(\mathbf{u}_{s-1}^{h_1}, p_{s-1}^{h_1})$. We set $(\mathbf{u}^{h_1}, p^{h_1}) = (\mathbf{u}_{\mathcal{S}}^{h_1}, p_{\mathcal{S}}^{h_1})$.

The general algorithm for a J-level method for approximating the solution of (6) can summarized as follows.

Step 1. Solve the following nonlinear system on the coarsest mesh h_0. Find $\mathbf{u}^{h_0} \in \mathbf{V}^{h_0}$ and $p^{h_0} \in S_0^{h_0}$ satisfying

$$\begin{aligned} a_0(\mathbf{u}^{h_0}, \mathbf{v}^h) + a_1(\mathbf{u}^{h_0}, \mathbf{u}^{h_0}, \mathbf{v}^h) + b(p^{h_0}, \mathbf{v}^h) &= (\mathbf{f}, \mathbf{v}^h) \quad \forall\, \mathbf{v}^h \in \mathbf{V}_0^{h_0}, \\ b(\mathbf{u}^{h_0}, \psi^h) &= -(g, \psi^h) \quad \forall\, \psi^h \in S^{h_0}, \\ \mathbf{u}^{h_0} &= \mathbf{q}^{h_0} \quad \text{on } \Gamma\,. \end{aligned}$$

Step 2. For each remaining mesh level $j = 1, \ldots, J$ solve the following linearized problems performing $\mathcal{S}$ Newton steps at each level.

- Set $\mathbf{u}_0^{h_j} = \mathbf{u}^{h_{j-1}}$.
- For $s = 1, 2, \ldots, \mathcal{S}$ solve the linear problem.
  Find $\mathbf{u}_s^{h_j} \in \mathbf{V}^{h_j}$ and $p_s^{h_j} \in S^{h_j}$ satisfying

$$\begin{aligned} a_0(\mathbf{u}_s^{h_j}, \mathbf{v}^h) + a_1(\mathbf{u}_{s-1}^{h_j}, \mathbf{u}_s^{h_j}, \mathbf{v}^h) + a_1(\mathbf{u}_s^{h_j}, \mathbf{u}_{s-1}^{h_j}, \mathbf{v}^h) + b(p_s^{h_j}, \mathbf{v}^h) \\ = (\mathbf{f}, \mathbf{v}^h) + a_1(\mathbf{u}_{s-1}^{h_j} \mathbf{u}_{s-1}^{h_j}, \mathbf{v}^h) \quad \forall\, \mathbf{v}^h \in \mathbf{V}_0^{h_j}, \end{aligned}$$

$$b(\mathbf{u}_s^{h_j}, \psi^h) = -(g, \psi^h) \quad \forall\, \psi^h \in S^{h_j}\,.$$

- Set $\mathbf{u}^{h_j} = \mathbf{u}_{\mathcal{S}}^{h_j}$, $p^{h_j} = p_{\mathcal{S}}^{h_j}$.

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In the algorithm J denotes the number of mesh levels and $\mathcal{S}$, typically one or two, the number of Newton steps at each level.

The question which automatically arises is how to choose the sequence of meshes so that we obtain optimal accuracy. This question was addressed from a theoretical standpoint in [6] and the results are summarized in the next section.

2.2. An error estimate. In [6], motivated by the work of Xu [10] on semilinear elliptic boundary value problems, a two-level discretization method was proposed for the incompressible Navier–Stokes equations with homogeneous Dirichlet boundary conditions. An estimate for the error in the fine-grid solution was also proven. A later paper [7] considered a multilevel discretization method for the Navier–Stokes equations. To motivate the choice of grid sizes in our experiments, we present a short summary of the error estimate presented in [6].

Consider a two-level method for the homogeneous Navier–Stokes equations with grids h_0 and h_1 where $h_0 > h_1$. We assume that the discrete problems (6) posed over the finite element spaces $(\mathbf{V}_0^{h_0}, S^{h_0})$ and $(\mathbf{V}_0^{h_1}, S^{h_1})$ have unique solutions given by $(\mathbf{u}^{h_0}, p^{h_0})$ and $(\mathbf{u}^{h_1}, p^{h_1})$, respectively, and that $(\mathbf{u}, p)$ is the unique solution to (2) with $\mathbf{q} = \mathbf{0}$ and $g = 0$. In this case, an error estimate of the form

$$
\begin{aligned}
|\mathbf{u} - \mathbf{u}^{h_1}|_1 + \|p - p^{h_1}\|_0 \quad \leq C \quad & \left(\inf_{\mathbf{v}^h \in \mathbf{V}_0^{h_1}} |\mathbf{u} - \mathbf{v}^h|_1 + \inf_{\psi^h \in S^{h_1}} \|p - \psi^h\|_0 \right) \\
& + C \, \|\mathbf{u} - \mathbf{u}^{h_0}\|_0^{1/2} \, |\mathbf{u} - \mathbf{u}^{h_0}|_1^{3/2}
\end{aligned} \tag{8}
$$

was proved in [6] for dimension $d = 3$. An estimate for the inhomogeneous Navier–Stokes equations can be obtained by handling the inhomogeneities in the usual manner; see [4].

As an application of this result suppose we choose finite element spaces $\mathbf{V}^{h_1}$ and S^{h_1}, such as the Taylor–Hood pair, such that

$$
\inf_{\mathbf{v}^h \in \mathbf{V}_0^{h_1}} |\mathbf{u} - \mathbf{v}^h|_1 + \inf_{\psi^h \in S^{h_1}} \|p - \psi^h\|_0 \leq C h_1^2 \|\mathbf{f}\|_0 \, .
$$

If the two-level discretization is used and if $h_0^\sigma \leq C h_1^2$, $\sigma = 9/2$, then (8) implies that

$$
|\mathbf{u} - \mathbf{u}^{h_1}|_1 + \|p - p^{h_1}\|_0 \leq C \|\mathbf{f}\|_0 h_1^2,
$$

which is optimal. By balancing error terms in an analogous way the widest possible scalings between coarse and fine meshes (i.e., fewest levels) of which still suffice for an optimally accurate approximation can be derived for the J-level method as well.

3. Numerical examples. This section presents the results of numerical experiments with the multilevel algorithm described in section 2. Specifically, we consider two examples: the first is a nonphysical example with a known exact solution and the second is the well-known driven cavity problem. We chose the later problem because there are numerous results in the literature available for comparison.

For both examples, the region Ω is the unit square $\{0 < x < 1, 0 < y < 1\}$. The finite element discretization uses a triangular mesh with the Taylor–Hood element pair which is known to satisfy the stability condition; see [5]. In the Taylor–Hood element, piecewise quadratic functions on triangles are used to approximate the velocity and piecewise linear functions on triangles to approximate the pressure.

In order to compare the efficiency of the proposed method all linear and nonlinear systems were solved in the same way. All nonlinear problems were solved by computing

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two simple iterations (see [4]) and then switching to Newton iterations until the norm of the difference in successive iterates was within a fixed tolerance. Simple iteration was used first since it is globally convergent in the case when the discrete problem has a unique solution. All linear equations were solved using the same banded LINPACK routine. When solving the linearized problem with a mesh spacing h_j, we need the solution $\mathbf{u}^{h_{j-1}}$ generated on a mesh with spacing h_{j-1}. To do this we interpolate the solution $\mathbf{u}^{h_{j-1}}$ onto the grid with spacing h_j. Also for the calculations presented here we chose $\mathcal{S} = 1$; previous computations indicate that this is adequate for the Taylor–Hood element.

For the first example, the exact solution is

$$\mathbf{u} = \left\{ \begin{array}{c} \sin(\pi x)\sin(2\pi x) \\ x^2(1-x)\sin(\pi y) \end{array} \right\},$$
$$p = (1 + y(y^2 - 4))\cos(\pi x)\,,$$

where the data $\mathbf{f}$, g, and $\mathbf{q}$ are appropriately prescribed. In this example the boundary data is homogeneous but $g \neq 0$. Our numerical results for this example are for $\mathcal{R}e = 1$. In all calculations for the first example we use a uniform grid spacing.

If we compute the solution to (6) using the usual finite element method employing the Taylor–Hood element, we expect the optimal rate of convergence for the velocity if $\mathbf{u} \in \mathbf{H}^3(\Omega)$ and $p \in H^2(\Omega)$; i.e.,

$$\|\mathbf{u} - \mathbf{u}^h\|_1 \leq Ch^2 \qquad \text{and} \qquad \|\mathbf{u} - \mathbf{u}^h\|_0 \leq Ch^3\,. \tag{9}$$

Since we have an exact solution for this example, we compute the experimental rate of convergence for the one-level method and verify that it agrees with the predicted rate. For the computations using a J-level scheme, we compute the experimentally observed rate of convergence and compare with that obtained by the standard one-level method.

In Table 1 we present the results obtained by using the usual finite element method; specifically we give the L^2- and H^1-errors and the experimentally observed rates of convergence. In all cases a total of four iterations were necessary to obtain the prescribed Newton tolerance. As seen from the table, the rates of convergence predicted by theory were obtained. For means of comparison with the multilevel discretization methods we give the CPU timing generated by performing the computations on a DEC Alpha 2100. We generated an estimate for the operation count, which is provided with the standard LINPACK banded solver. In particular, we used the operation count $2nm^2$ for factoring an $n \times n$ matrix with a half bandwidth m. We note that timing data is highly machine and implementation dependent. However, since we used essentially the same implementation for the usual and multilevel discretizations and the same machine, it is no surprise that the relative timing data is consistent with the operation count.

In Table 2 we present results for the first example using a two-level method, and in Table 3 we present results using three- and four-level methods. In presenting these computations, we fix the coarse grid h_0 and then vary the fine grid. The rate of convergence was obtained by comparing the error on two consecutive fine grids which used the same initial coarse grid. From the error estimate (8) for a two-level method, we see that when the Taylor–Hood element is used we can choose the fine mesh as fine as $h_1 \sim h_0^{9/4}$ or, equivalently, the coarse mesh as coarse as $h_0 \sim h_1^{4/9}$ in order to balance the errors in the right-hand side of (8) and obtain the desired optimal rate

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TABLE 1
Results for the first example using a standard one-level finite element method.

h	Number of operations	CPU time	L^2-error	L^2-rate	H^1-error	H^1-rate
1/2	4.860×10^4	0.11	0.1028×10^0		1.8040	
1/4	1.205×10^6	0.68	0.1756×10^{-1}	2.54	0.5077	1.82
1/8	2.141×10^7	8.99	0.1948×10^{-2}	3.17	0.1241	2.03
1/16	3.617×10^8	151.0	0.2353×10^{-3}	3.05	0.0308	2.01
1/3	3.569×10^5	0.28	0.4535×10^{-1}		0.9153	
1/6	6.539×10^6	2.95	0.4776×10^{-2}	3.25	0.2221	2.04
1/9	3.471×10^7	14.6	0.1355×10^{-2}	3.11	0.0979	2.02
1/18	5.829×10^8	243.5	0.1649×10^{-3}	3.04	0.0243	2.01
1/5	3.067×10^6	1.49	0.8522×10^{-2}		0.3217	
1/10	5.342×10^7	22.2	0.9813×10^{-3}	3.12	0.0792	2.02
1/20	8.928×10^8	372.9	0.1200×10^{-3}	3.03	0.0197	2.01

TABLE 2
Results for the first example using a two-level method.

h_0	h_1	Number of operations	CPU time	L^2-error	L^2-rate	H^1-error	H^1-rate
1/2	1/4	3.499500×10^5	0.24	0.1760×10^{-1}		0.5080	
1/2	1/8	5.402142×10^6	2.36	0.1969×10^{-2}	3.16	0.1242	2.03
1/2	1/16	9.047408×10^7	38.3	0.3298×10^{-3}	2.58	0.0311	2.00
1/3	1/6	1.991766×10^6	1.01	0.4775×10^{-2}		0.2221	
1/3	1/9	9.035328×10^6	3.96	0.1355×10^{-2}	3.11	0.0979	2.02
1/3	1/18	1.460848×10^8	61.7	0.1709×10^{-3}	2.99	0.0244	2.01
1/4	1/8	6.556942×10^6	3.01	0.1948×10^{-2}		0.1241	
1/4	1/16	9.163088×10^7	39.2	0.2354×10^{-3}	3.05	0.0308	2.01
1/8	1/16	1.118396×10^8	47.1	0.2353×10^{-3}		0.0308	

TABLE 3
Results for the first example using three- and four-level methods.

h_0	h_1	h_2	h_3	Number of operations	CPU time	L^2-error	L^2-rate	H^1-error	H^1-rate
1/2	1/4	1/8		5.7×10^6	6.483	0.1948×10^{-2}		0.1241	
1/2	1/4	1/16		9.1×10^7	29.959	0.2353×10^{-3}	3.05	0.0308	2.01
1/4	1/8	1/16		9.7×10^7	36.817	0.2353×10^{-3}		0.0308	
1/2	1/4	1/8	1/16	9.6×10^7	26.313	0.2353×10^{-3}		0.0308	

of convergence. The experimental rates of convergence for the two-level method are optimal for our given choice of grids except for the L^2-rates where we used the very coarse initial mesh of $h_0 = 0.5$. Also, in this case when $h_1 = 1/8$, $h_1^{4/9} = 0.40$ so that a choice of $h_0 = .5$ may not be fine enough; similarly for $h_1 = 1/16$, $h_1^{4/9} = 0.29$ so that a choice of $h_0 = .5$ also may not be adequate. In all other computations h_0 was chosen so that $h_0 \le h_1^{4/9}$.

We want to compare the accuracy and relative efficiency of the standard Galerkin method with the multilevel method. To do this, we compare the solution generated at a fixed value of h for the one-level method with the results obtained using a multilevel method where the finest grid has the same mesh spacing h. For example, for $h = 1/8$ in the one-level method we can compare this with our results using a two-level

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method with $h_0 = 1/2, h_1 = 1/8$ and one with $h_0 = 1/4, h_1 = 1/8$; we can also compare with a three-level method with $h_0 = 1/2, h_1 = 1/4, h_3 = 1/8$. Table 4 summarizes these results by combining the results from Tables 1–3. In Table 4 if only one grid is indicated then the results correspond to the standard one-level finite element method. In all two-level computations when h_0 was chosen so that $h_0 \leq h_1^{4/9}$ we observed the optimal predicted rate of convergence and the method was three to four times faster than the one-level method when we compare CPU times and number of operations. Also, the finer grid calculations gave the greater savings in time and number of operations. In most cases a three- or four-level method was comparable in efficiency with a two-level approach. Thus most of the savings for two-dimensional problems at least were already obtained by using two levels.

TABLE 4
Comparison of multilevel methods to a one-level method for the first example.

Grids	Total no. of operations	CPU time	L^2-error	H^1-error
1/4	1.2054×10^6	0.68	0.1756×10^{-1}	0.5077
1/4, 1/2	3.500×10^5	0.24	0.1760×10^{-1}	0.5080
1/8	2.141×10^7	8.99	0.1948×10^{-2}	0.1241
1/8, 1/4	6.559×10^6	3.01	0.1948×10^{-2}	0.1241
1/8, 1/2	5.402×10^6	2.36	0.1969×10^{-2}	0.1242
1/8, 1/4, 1/2	5.703×10^6	2.56	0.1948×10^{-2}	0.1241
1/16	3.617×10^8	150.98	0.2353×10^{-3}	0.0308
1/16, 1/8	1.118×10^8	47.09	0.2353×10^{-3}	0.0308
1/16, 1/4	9.163×10^7	39.20	0.2354×10^{-3}	0.0308
1/16, 1/2	9.047×10^7	38.26	0.3298×10^{-3}	0.0311
1/16, 1/8, 1/4	9.698×10^7	41.75	0.2353×10^{-3}	0.0308
1/16, 1/4, 1/2	9.078×10^7	38.79	0.2353×10^{-3}	0.03081
1/16, 1/8, 1/4, 1/2	9.614×10^7	41.41	0.2353×10^{-3}	0.0308
1/6	6.540×10^6	2.95	0.4776×10^{-2}	0.2221
1/6, 1/3	1.992×10^6	1.01	0.4775×10^{-2}	0.2221
1/9	3.471×10^7	14.57	0.1355×10^{-2}	0.0979
1/9, 1/3	9.035×10^6	3.96	0.1355×10^{-2}	0.0979
1/18	5.829×10^6	243.51	0.1649×10^{-3}	0.0243
1/18, 1/3	1.461×10^8	61.74	0.1709×10^{-3}	0.0244

The second example is the driven cavity flow problem on the unit square with homogeneous boundary conditions on three sides, and on the top we set the tangential velocity to be one and the normal velocity to be zero. Numerical computations were performed on a triangular mesh with the Taylor–Hood element pair and a nonuniform grid. The nonuniform mesh in the x-direction was determined by mapping a uniform grid by the function $\sin^2(\pi x/2)$; the nonuniform grid in the y-direction was determined analogously. This nonuniform grid allows better resolution at the corners. In computing the solution at higher values of the Reynolds number, we used a continuation method.

For the driven cavity problem, we compute an approximate solution for $\mathcal{R}e = 100$, $\mathcal{R}e = 400$, and $\mathcal{R}e = 1000$ on various grids using the standard one-level method and multilevel methods. We compare these results with those published by Ghia, Ghia, and Shin [2] where a coupled, strongly implicit multigrid method is used with 129 grid points in each direction. In particular, we plot the horizontal component of the velocity along the vertical line passing through the geometrical center of the cavity and the vertical component of the velocity along the horizontal line passing through

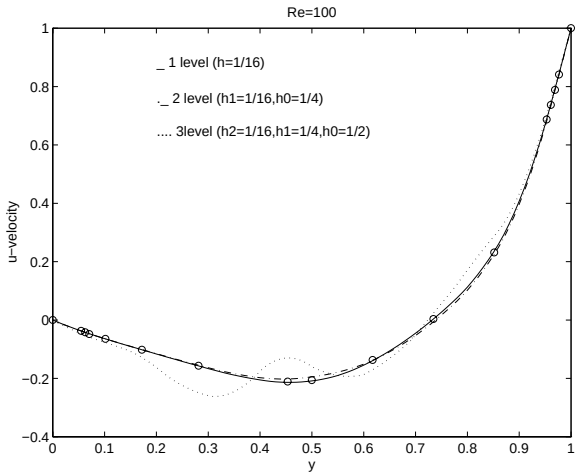

FIG. 1. *Comparison of u-velocity for Reynolds number*= 100.

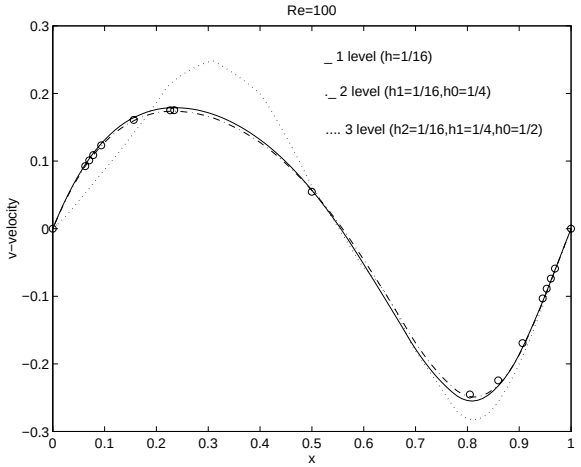

FIG. 2. *Comparison of v-velocity for Reynolds number*= 100.

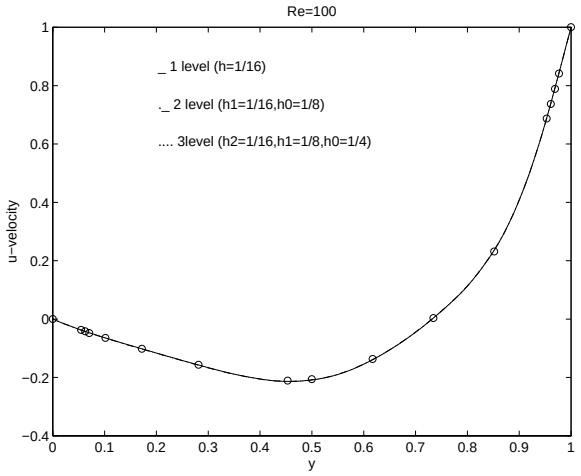

FIG. 3. *Comparison of u-velocity for Reynolds number*= 100.

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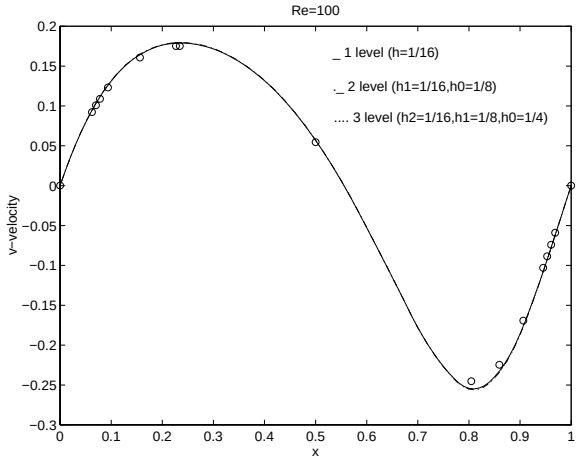

FIG. 4. *Comparison of v-velocity for Reynolds number*= 100.

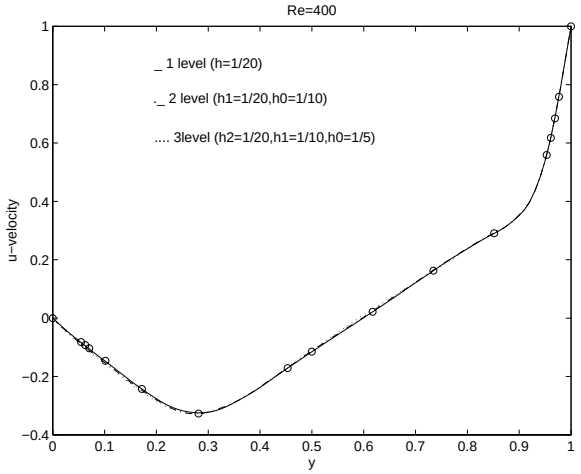

FIG. 5. *Comparison of u-velocity for Reynolds number*= 400.

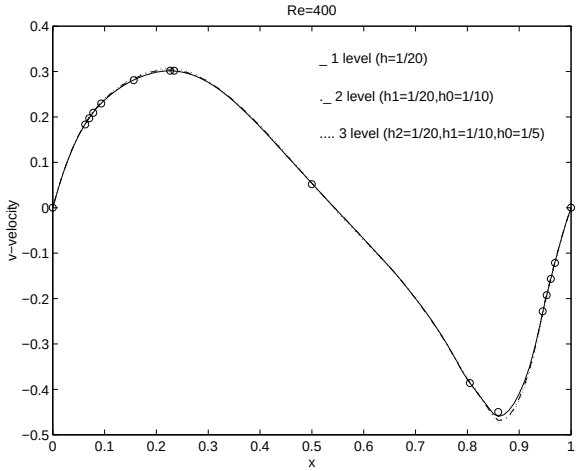

FIG. 6. *Comparison of v-velocity for Reynolds number*= 400.

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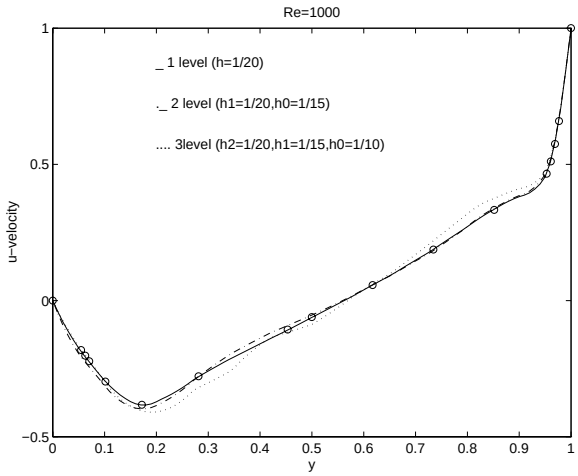

FIG. 7. *Comparison of u-velocity for Reynolds number*= 1000.

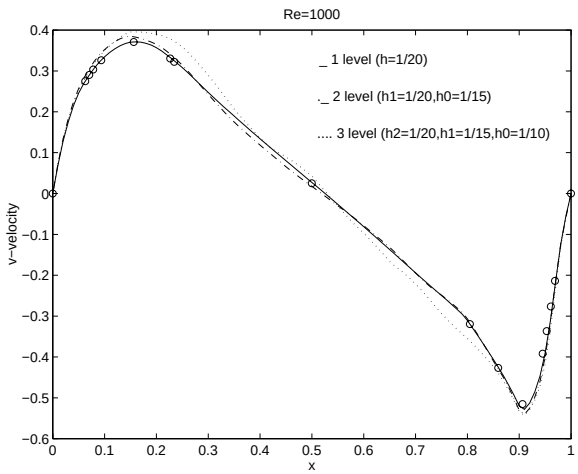

FIG. 8. *Comparison of v-velocity for Reynolds number*= 1000.

the geometrical center of the cavity. In all figures the small circles correspond to the published results of Ghia, Ghia, and Shin, and the solid line corresponds to the standard one-level finite element method. In all figures when we give h we are setting the value h would be if a uniform mesh were used.

Figures 1–4 correspond to results computed for Reynolds number 100 for a fine grid of $h = 1/16$. In comparing the results from the two-level method with those from the one-level method, we see from Figures 1–2 that the results are quite close when the coarse mesh $h_0 = 1/4$; from Figures 3–4 we see that when the coarse mesh $h_0 = 1/8$ the results are indistinguishable from the one-level method. Note from Figures 1–2 that the results from the three-level method where the nonlinear problem is solved on the coarse mesh $h_0 = 1/2$ are unsatisfactory. This is because the nonlinear solution on the coarse grid is not accurate enough. Note that in Figures 3–4 where we use a coarse mesh of $h_0 = 1/4$ for the three-level method the results are indistinguishable from the one-level result.

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For Reynolds number equal to 400 and 1000, we present results using a finest mesh size of $h = 1/20$ in Figures 5–6 and Figures 7–8, respectively. At $\mathcal{R}e = 400$ we see that the two- and three-level methods are again indistinguishable from the standard finite element method and the results of [5]. For $\mathcal{R}e = 1000$ the three-level method is again beginning to provide unsatisfactory results. This is because the original coarse grid is too coarse to capture the envelope of the solution with the usual Galerkin discretization; with $\mathcal{R}e = 1000$ and $h_0 = 1/10$ the cell Reynolds number is just too large. If one still desires to compute approximate solutions at higher Reynolds number using very coarse meshes, then either a stabilized variational formulation should be used on the coarse mesh (e.g., upwind Galerkin) or on some sort of subgrid-scale model. However, substantial savings can still be achieved by a two-level approach. For example, if we compare the operation count for solving a nonlinear system for $h = 1/20$ with solving a nonlinear system for $h = 1/15$ and a linear system for $h = 1/20$ we see that considerable savings are obtained. Using the operation count $2nm^2$ and assuming five Newton iterations to solve the nonlinear problems, we see that the nonlinear system for $h = 1/20$ requires about 1.12×10^9, whereas the nonlinear system for $h = 1/15$ plus one linear system at $h = 1/20$ requires 5.66×10^8 operations.

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